THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** phases	Illuminated part changes
2	**(c)** new moon	Dark side faces us
3	**(a)** does not change	Only visible portion changes
4	**(a)** Moon's phases and the seasons	Adjusts both
5	**(d)** 11 days	365 − 354
6	**(c)** 2000	Divisible by 400
7	**(a)** sidereal year	Slightly longer than tropical year
8	**(c)** 21 March 1956	1 Chaitra 1878 Saka
9	**(b)** solstices	Around June 21 and December 21
10	**(a)** useless human-made objects in orbit	Old satellites, rocket parts

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Changing shape of the illuminated part seen from Earth (1). Occurs over nearly 29.5 days (1).	1+1
12	We see different parts of its lit half (1). The Moon itself does not change (1).	1+1
13	New moon (1). Full moon (1).	1+1
14	Lunar follows the Moon's phases (1). Solar follows the seasons (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	New moon to waxing crescent to first half (1). Gibbous to full moon (1). Waning back to new moon (1).	1+1+1
16	Moon moves in orbit around Earth (1½). The lit part visible to us decreases (1½).	1½+1½
17	Noticed phases repeat every 29.5 days (1). Month defined by cycle (1). 12 months made the lunar year (1).	1+1+1
	OR: Earth takes about 365¼ days (1). Extra day every four years (1). Added to February (1).	1+1+1
18	Earth takes slightly less than 365¼ days (1). Skipping century years corrects it (1). Years divisible by 400 are leap years (1).	1+1+1
	OR: Uses Moon's phases for months (1). Adds extra months to match the seasons (1). Traditional Indian calendars (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Moon orbits Earth (1). Half always lit by Sun (1). We see different portions (1). Phases from new moon to full moon (1). Cycle of 29.5 days (1).	1+1+1+1+1
	OR: Lunar: Moon's cycles, year 354 days (1½). Solar: seasons, 365 days with leap years (1½). Luni-solar: Moon months plus extra month (2).	1½+1½+2
20	Divisible by 4 is leap (1). Century years skipped unless divisible by 400 (1). 1900 not a leap year (1). 2000 leap year (1). Reason: Earth takes slightly less than 365¼ days (1).	1+1+1+1+1
	OR: Begins 22 March (1). Months 30 or 31 days (1). Leap year by adding a day to Chaitra (1). Adopted 1956 (1). Follows Surya Siddhanta principles (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 11: **Keeping Time with the Skies** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 29.5 days	Nearly one month
2	**(d)** full moon	Purnima
3	**(d)** 50 minutes	Moon moves along its orbit
4	**(a)** 365¼ days	One year
5	**(c)** Adhika Maasa	Intercalary month
6	**(a)** 1900	Century not divisible by 400
7	**(b)** 20 minutes	A very small difference
8	**(a)** solar calendar	Almost the same Gregorian date each year
9	**(b)** a human-made object orbiting Earth	Launched by rockets
10	**(c)** Vikram Sarabhai	Founder of ISRO's early efforts

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A year with an extra day in February (1). Earth takes about 365¼ days to orbit the Sun (1).	1+1
12	12 lunar months add up to 354 days (1). Solar year is about 365 days (1).	1+1
13	An extra month added in some calendars (1). It fixes the gap between lunar and solar years (1).	1+1
14	22 March (1). 21 March in leap years (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lunar year 354 days (1). Solar year about 365 days (1). Extra month added (1).	1+1+1
16	Begins 22 March with Chaitra (1). 365 days with months of 30 or 31 days (1). Adopted from 21 March 1956 (1).	1+1+1
17	They follow the lunar calendar of 354 days (1½). So they shift about 11 days earlier every year (1½).	1½+1½
	OR: Sun rises exactly east on equinoxes (1). Extremes on solstices (1). Path observation gives a year of about 365 days (1).	1+1+1
18	Human-made objects orbiting Earth (1). Weather and mapping (1). Communication and research (1).	1+1+1
	OR: Cartosat – mapping (1). AstroSat – astronomy (1). AzaadiSat or InspireSat-1 – student or small satellite (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sunrise positions vary across the year (1). Equinoxes and solstices (1). Star patterns at sunset (1). Year of about 365 days (1). Surya Siddhanta record (1).	1+1+1+1+1
	OR: Tropical year equinox to equinox (1½). Sidereal year same star at sunset (1½). Differ by 20 minutes (1). Seasons shift slowly if sidereal is used (1).	1½+1½+1+1
20	Solar sidereal festivals fixed, e.g. Pongal (2). Lunar festivals shift (2). Reason: lunar year 354 days (1).	2+2+1
	OR: Launched by rockets (1). Communication (1). Earth observation (1). Astronomy (1). Examples Cartosat, AstroSat (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.